

System Architecture — Whole System Class Diagram

Aligned with SYSTEM_ARCHITECTURE.md · 3-Tier · MVC · All Modules · Billing & Reports

■ Implemented ✓

■ Planned →

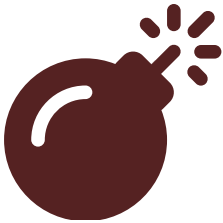
UML: namespace = layer/module · ..► dependency · —► association

Figure 1 — Three-Tier Architecture (Class View) · SYSTEM_ARCHITECTURE §1



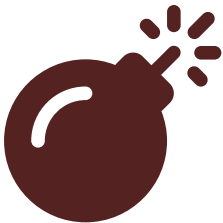
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Figure 2 — Module Architecture (All Modules → Classes) · §3 & §12



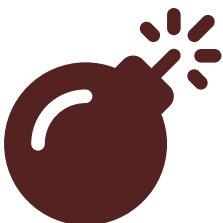
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Figure 3 — MVC Request Flow (Class Interactions) · §4



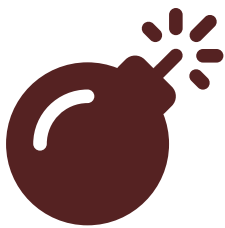
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Figure 4 — Security & RBAC Class Diagram · §5



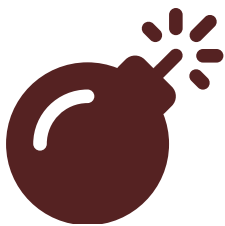
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Figure 5 — Domain Model (Implemented + Planned) · \$6



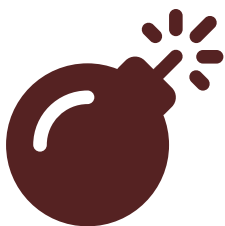
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Figure 6 — Billing Module Classes  · \$10



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Figure 7 — Reports Module Classes  · \$11



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Figure 8 — Complete Class Inventory

Layer	Implemented 	Planned 
Models	User, Order, CatalogDesign, CatalogImage, ProductionLog, MetalPrice	Invoice, InvoiceItem, Payment, PaymentMethod, Category, Notification, ReportExport
Controllers	Auth (8), Customer (2), Admin (8), Technician (2), Catalog, Profile	InvoiceController, PaymentController, ReportController
Middleware	EnsureCustomer, EnsureAdmin, EnsureTechnician, EnsurePermission	—
Enums	UserRole, Permission, OrderStatus, DesignType, AvailabilityStatus	—
Support	Rbac, ValidationRules, ValidPhone	InvoiceService, ReportEngine
Form Requests	22 classes	StoreInvoiceRequest, StorePaymentRequest, FilterReportRequest
View Components	AppLayout, AdminLayout, TechnicianLayout, GuestLayout, PublicLayout	—

Source: docs/SYSTEM_ARCHITECTURE.md · docs/SYSTEM_ARCHITECTURE_CLASS_DIAGRAM.md

Print → **PDF** for Word/project report.

Architecture pattern: 3-Tier MVC — Browser → Laravel Controllers → Eloquent Models → MySQL + File Storage.